DS4420
Project Phase II
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GitHub: https://github.com/anc002/flight-delay-prediction

I. Introduction & Literature Review

Flight delays cost the U.S. aviation industry billions of dollars each year and disrupt travel plans of millions of passengers. Because aircraft crews and gates serve multiple flights in sequence, a single delayed departure can cascade through an entire day of connected routes. Predicting whether a flight will depart at least 15 minutes behind schedule, before it actually departs, gives airlines an opportunity to reallocate resources and gives passengers earlier warning of potential disruptions.

This project has the following research question: *Can we predict whether a U.S. domestic flight will be delayed by 15 or more minutes before departure, and what factors drive that risk?*

Two methods are applied. First is a Multi Layer Perceptron (MLP) which is implemented in Python using Keras. The second is a Bayesian logistic regression model which is manually implemented in R using a Metropolis Sampler. The MLP learns non-linear relationships between schedule-based input features and delay risk. The Bayesian model returns posterior distributions over each coefficient, therefore providing uncertainty estimates which the MLP cannot produce.

The target variable is DEP_DELAY15, a binary indicator equal to 1 when a flight's actual departure is 15 or more minutes later than its scheduled departure time. This variable was chose over arrival delay to avoid using any information that only becomes available after the flight has already left the gat, such as elapsed flight time or actual arrival time.

Literature Review

Rebello and Balakrishnan (2014) predict departure delays 2 to 24 hours in advance using operational data from the U.S. National Airspace System. Their model incorporates time-based variables along with network-level congestion conditions across airports and finds that both temporal features and upstream delay propagation are strong predictors. The features used in this project, , especially departure hour and carrier, are in line with the variables their work identify to be informative.

Hatipoglu and Tosun (2024) apply several ML methods to flight delay classification using operational airline data. Their feature set includes time of day, day of week and network conditions at connected airports. Their results confirm that temporal variables reliably support delay classification which is consisten with the EDA patters discussed later in the report.

Qu et al. (2020) train deep convolutional neural networks on approximately four million flight records, combining schedule data with weather observations at the departure airport. Their results show that adding weather substantially improves prediction accuracy over models only using schedule data. This project does not include weather data. The performance reported here therefore represent a ceiling that schedule based features can achieve, which the literature suggests falls in the 0.70 to 0.75 ROC-AUC range before adding weather.

Bayesian approaches to classification offer a practical advantage over point-estimate methods. A standard classifier returns a single predicted probability over each flight. A Bayesian model returns a full posterior distribution over each coefficient, allowing how confident a model is in any given effect to be quantified. In the context of operations or scheduling, this uncertainty estimate supports more informed decisions than a point prediction alone.

II. Methodology and Data Collection

Data Collection and Cleaning

The dataset comes from the [Bureau of Transportation Statistics (BTS) ON-Time Perfromance database](), which covers all U.S. flights from January through December 2025. Each month was downloaded separately (limitation of the website) and compiled into one master dataset manually. The raw file contains ~ 7 million rows. Cancelled flights and diverted flights (~ 122k rows) were removed first as neither can receive a departure delay. Rows where DEP_DEL15 were missing was then dropped (~98k rows). One additional row with an invalid scheduled departure time was removed. The cleaned dataset contained 6,879,484 flights, of which 21.7% were delayed and 78.3% were on time.

The scheduled departure time column CRS_DEP_TIME was converted to a departure hour feature (DEP_HOUR), an integer from 0 to 23 and the original column was dropped. The final dataset has nine columns: MONTH, DAY, DAY_OF_MONTH, DAY_OF_WEEK, OP_UNIQUE_CARRIER, ORIGIN, DEST, DEP_DEL15, DISTANCE and DEP_HOUR

Exploratory Data Analysis

Several patterns in the data shaped the decisions described later in this section

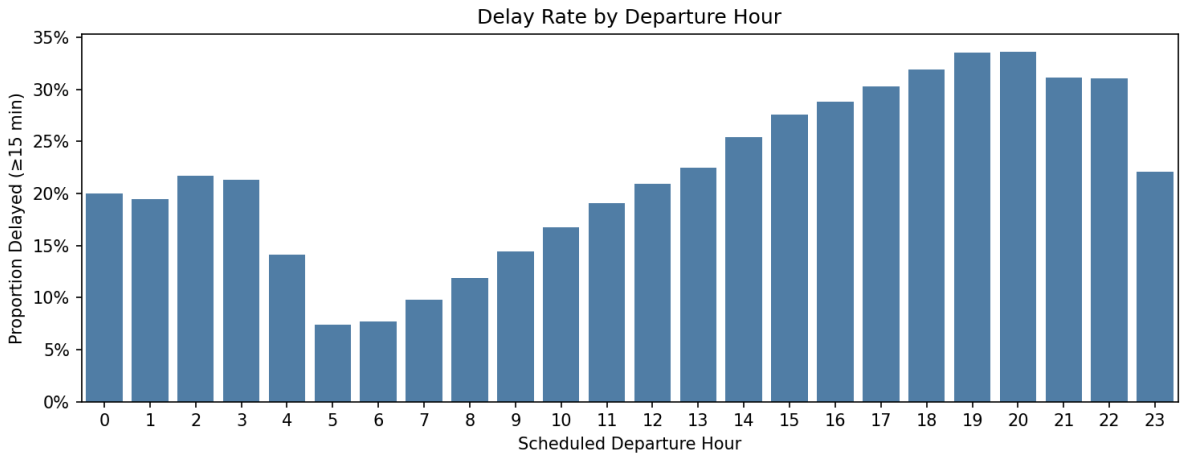

Figure 1: Delay Rate by Departure Hour

Figure 1 shows delay rate by scheduled departure hour. FLights at hours 0 through 3 have delay rates between 19% and 22%, reflecting overnight departures carry forward accumulated delays over the previous day. Rates drop sharply to a minimum of around 7% at hour 5, when frest aircraft and crews begin the day's first lot of flights. From there, delay rates rise steadily through the afternoon and

evening, peaking at approximately 33% at hours 19 and 20 before falling back to around 22% at hour 23. This non-linear pattern justifies using DEP_HOUR as a numeric feature and makes it the strongest predictive signal in both models

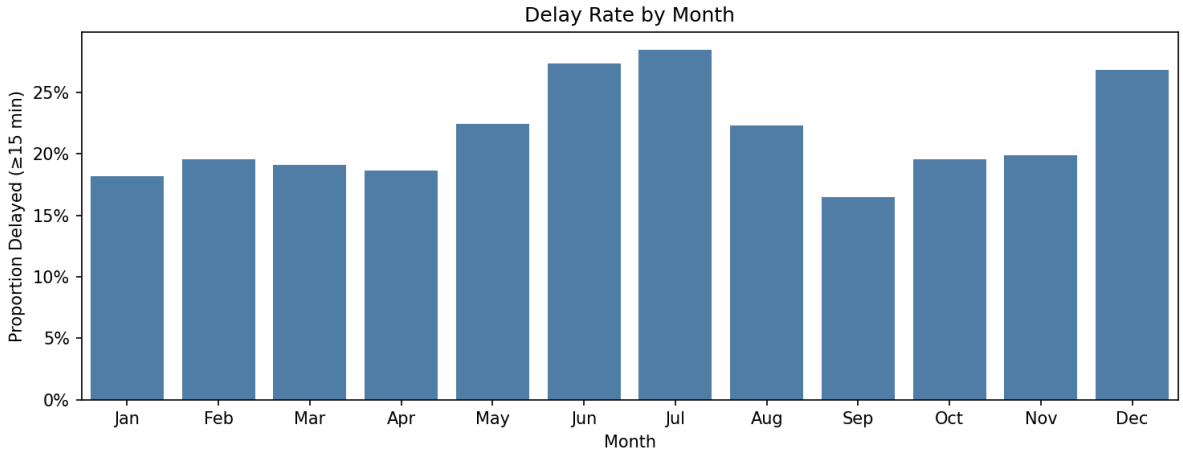

Figure 2: Delay Rate by Month

Figure 2 shows delay rate by month. September is the lowest month at approximately 16% with July having the highest at approximately 28%, followed by June and December at around 27%. The remaining months are around 18-19%. The range across months spans roughly 12 pct points. Because the relationship between month and delay rates do not have a meaningful linear ordering, for example, December is not “more” than June in a numeric sense, MONTH is one-hot coded in both models instead of being treated as a scaled integer.

Additional EDA found that Sunday and Monday have the highest delay rates by day of week at roughly 25% and 23% respectively, with Tuesday the lowest at around 13%. Among the 14 carriers Frontier, PSA and JetBlue show the highest delay rates at approximately 27%, 26% and 25% respectively. Among airports with at least 5,000 departures, ASE, SFB and DFW have the highest rates. Flight distance shows almost no variation in delay rate across distance buckets, ranging only from about 20% to 23%, which provides an early indication that distance contributes limited predictive value.

Subsampling

A stratified 600,000 row subsample was drawn from the cleaned data for the MLP and a further stratified 50,000 row subsample was drawn from that for the Bayesian model to account for computational limitations. Both subsamples stratify jointly on MONTH and DEP_DEL15, ensuring all 12 months are proportionally represented and the 78/22 class split is preserved in the samples.

To limit the number of encoded columns, airports outside the top-N by flight count are grouped as "Other." The MLP uses the top 50 airports, which account for 78.4% of flights, while the Bayesian model uses the top 20, which account for 51.8% of flights, to further reduce the parameter space given MCMC scaling constraints.

Multi Layer Perceptron (MLP)

A Multi-Layer Perceptron extends linear regression by adding weight matrices and activation functions, allowing it to capture non-linear relationships between features and the output. The network has two hidden layers of width 64 and 32, with Dropout at rate 0.3 after each. The forward pass computes:

$$h^{(1)} = \mathrm{ReLU}(W^{(1)} x)$$
$$h^{(2)} = \mathrm{ReLU}(W^{(2)} h^{(1)})$$
$$f(x) = \mathrm{sigmoid}(W^{(3)} h^{(2)})$$

The weight matrices are learned by minimizing binary cross-entropy using the Adam optimizer over 50 epochs with batch size 128. Due to the 78/22 class imbalance, class weights of 0.64 (on-time) and 2.31 (delayed) are applied so the model is penalized more heavily for missing a delay. An 80/20 stratified train-test split gives 480,000 training rows and 133 features after one-hot encoding categorical variables and standardizing numeric ones.

Bayesian Logistic Regression

Bayesian logistic regression is a probabilistic approach to classification that, rather than returning a single set of weights, estimates a full posterior distribution over the weight vector w. Each observation follows a Bernoulli data likelihood with probability given by the sigmoid function::

$$y_i \mid w, x_i \sim \mathrm{Bernoulli}(\mathrm{sigmoid}(w^T x_i))$$

A weakly informative Normal prior N(0, 10) is placed on each coefficient. Because the posterior has no closed form, it is sampled using a Metropolis algorithm with an isotropic Gaussian proposal, where the acceptance ratio simplifies to:

$$\alpha = p(w' \mid y, X) / p(w_0 \mid y, X)$$

sigma = 0.01 was selected after trial runs, yielding a 27% acceptance rate within the 20-40% target range. The sampler ran for 100,000 iterations, with 18% burn-in and thinning by 50, giving 1,640 approximately independent posterior samples. The design matrix has 74 columns after one-hot encoding with top-20 airport grouping.

III. ML Results and Interpretations

MLP Results

The MLP achieved a test ROC-AUC of 0.7154 and test accuracy of 66.99%. Accuracy alone is not a reliable metric here because a model that predicts on-time for every flight would achieve 78% accuracy while identifying zero delays. ROC-AUC measures the probability that the model ranks a randomly chosen delayed flight above a randomly chosen on-time flight across all possible decision thresholds, making it the more meaningful metric for an imbalanced classification problem.
At a 0.5 decision threshold, the confusion matrix shows 63,866 true negatives, 16,519 true positives, 9,514 false negatives, and 30,101 false positives. Recall on the delayed class is 0.63, meaning the

model correctly identifies 63% of actual delays. Precision on the delayed class is 0.35, meaning that when the model predicts a delay, it is correct 35% of the time. The relatively high false positive rate follows directly from the class weights applied during training: the model is deliberately tuned to prioritize catching delays over avoiding false alarms, which is the appropriate tradeoff in a scheduling context where a missed delay is more costly than an unnecessary alert.

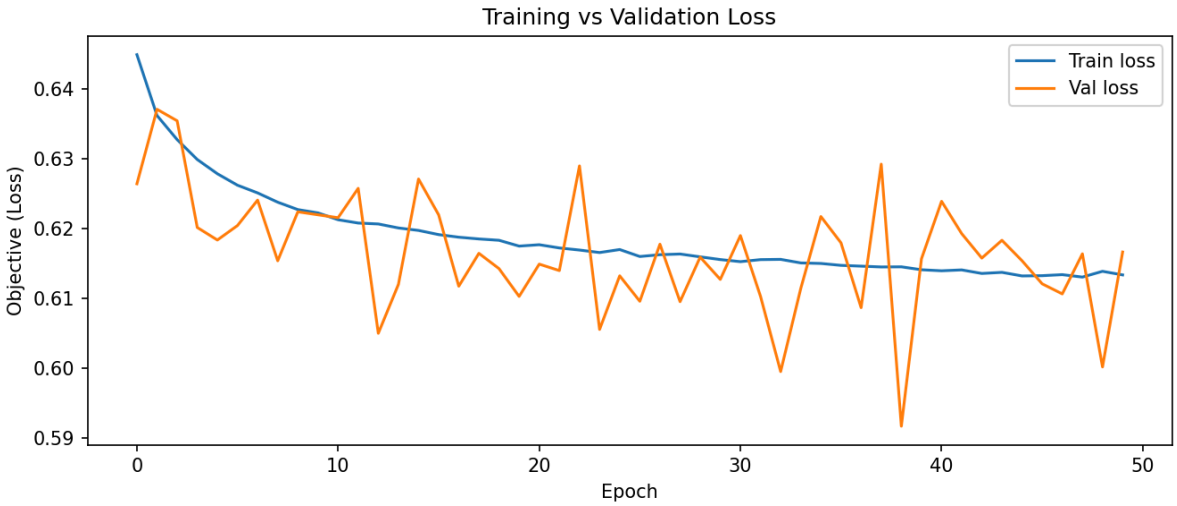

Figure 3: MLP Training vs. Validation Loss

Figure 3 shows the binary cross-entropy objective function on the training and validation sets across all 50 epochs. Training loss decreases steadily throughout. The validation loss is noisier, which is expected given the smaller validation set size, but follows a downward trend overall and does not rise consistently above the training loss. This indicates that dropout regularization was effective and the model did not overfit. The training curve flattens after approximately epoch 30, suggesting the model approached its performance ceiling with the current feature set.

Bayesian Logistic Regression Results

The Bayesian model achieved a ROC-AUC of 0.6694, computed by applying the sigmoid function to X_test * w_hat, where w_hat is the vector of posterior mean coefficients from the 1,640 thinned samples.

Convergence diagnostics confirm the sampler explored the posterior adequately. Trace plots for DEP_HOUR and DISTANCE show good mixing throughout, with no extended drifts or periods where the chain appears stuck. The Alaska Airlines carrier trace shows somewhat slower mixing, which is consistent with a weaker and more uncertain signal in the data. ACF plots for DEP_HOUR and DISTANCE drop to near zero within the first lag after thinning, confirming the thinned samples are approximately independent, however, Alaska Airlines drops off more slowly, showing a potential lack of independence.

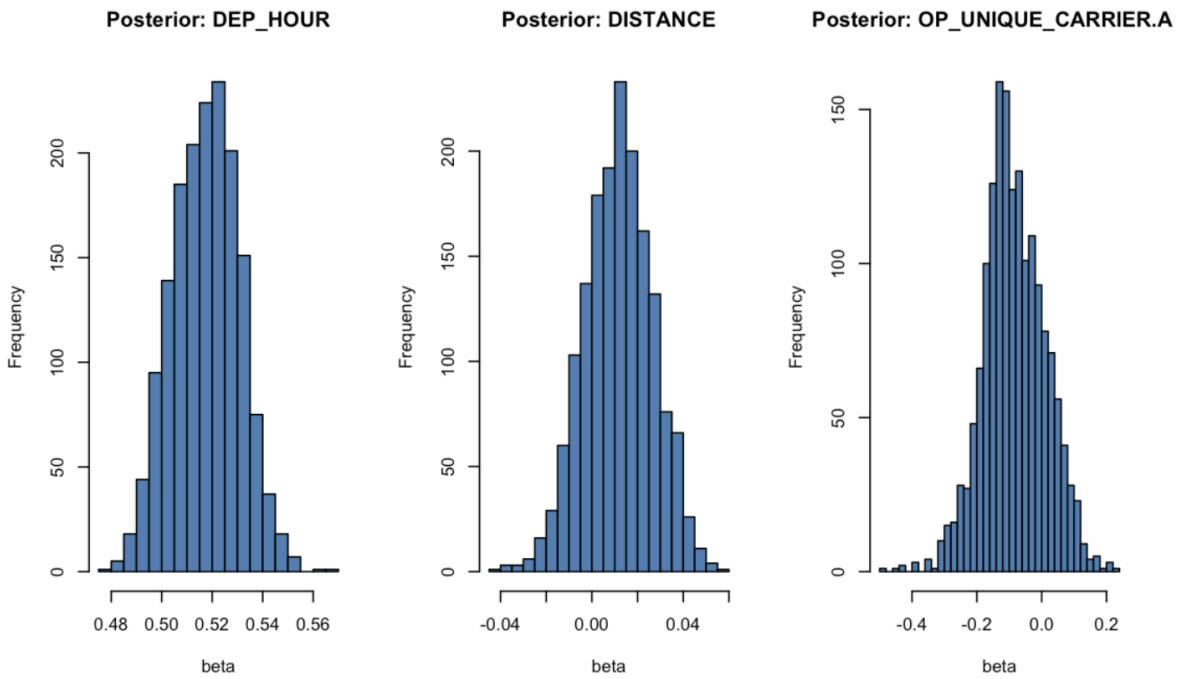

Figure 4: Posterior Distributions for DEP_HOUR, DISTANCE, and Alaska Airlines Coefficient

Figure 4 shows the posterior distributions for DEP_HOUR, DISTANCE, and the Alaska Airlines carrier indicator. The 95% credible intervals for these coefficients are:

- DEP_HOUR: [0.49, 0.54]. The entire interval lies above zero. Later departures are reliably associated with higher delay risk, consistent with delay accumulation across a day's schedule.
- DISTANCE: [-0.018, 0.040]. The interval overlaps zero. Flight distance does not have a reliably signed effect on departure delay probability once temporal features are accounted for. This aligns with the EDA finding that delay rates are nearly flat across distance buckets, spanning only 20% to 23%.
- Alaska Airlines (AS): [-0.28, 0.11]. The interval overlaps zero. Carrier is one-hot encoded, meaning each airline becomes its own binary column. For carriers with relatively few flights in the 50,000-row subsample, these columns are sparse, giving the sampler less information to work with and producing wider, less certain posteriors. Individual carrier effects are generally weaker than temporal effects in this feature set.

Table 1: Model Comparison

Metric	MLP	Bayesian Log Reg
ROC-AUC	0.7154	0.6694
Recall	0.63	N/A
Precision	0.35	N/A
Training Observations	480,000	50,000

IV. Conclusions and Future Work

This project shows that schedule-based features can predict U.S. domestic departure delays with moderate reliability. The MLP achieved a ROC-AUC of 0.7154 and the Bayesian logistic regression achieved 0.6694, placing both models in the range the literature associates with schedule only prediction. Qu et al. (2020) show that adding weather data substantially improves accuracy beyond this range and these results are consistent with that finding.

Departure hour is the strongest predictor in both models. This is confirmed by the Bayesian model shown by the tight posterior entirely above zero with a 95% credible interval of [0.49, 0.54]. Distance shows almost no predictive value, which is supported by both the flat EDA pattern across distance buckets and a credible interval that overlaps zero. Carrier effects are present in the data but are relatively weak compared to temporal features. Sparse one-hot encoded columns for smaller carriers also contribute additional uncertainty in the Bayesian model.

The Bayesian model assumes a Bernoulli likelihood with a signmoid link and independent Normal priors N(0,10) on all coefficients. These priors are weakly informative and place no directional belief on any feature. The Metropolis sampler assumes the isotropic Gaussian proposal is appropriately scaled which is supported by the 27% acceptance rate within the 20 to 40% target range. Trace plots and ACF plots confirm adequate mixing and near zero autocorrelation for most features after thinning. The MLP assumes the chosen architecture, dropout rate and class weights are sufficient to learn generalisable patterns. The convergence of training and validation loss supports this.

The most significant limitation is the absence of weather data. Departure delays are frequently caused by conditions at the origin airport that no schedule feature can capture. Another limitation is the absence of prior flight delay. A flight is more likely to be delayed if the same aircraft arrived late on its previous leg. This feature could be constructed through tail number joins but was not included. For the bayesian model, subsampling to 50,000 rows limits exposure to rare airport and carrier combinations (such as Alaska Airlines), reducing reliability for those specific coefficients.

To extend this project, the most impactful addition would be NOAA weather observations at departure airports. Encoding prior-flight delay via tail number joins would address operational propagation effects. On the modelling side, a more scalable sampling method would allow the Bayesian model to train on the full dataset without subsampling, producing more reliable posterior for all 74 coefficients.

V. Appendix

MLP Code

```
n_features = X_train.shape[1]

inpx    = Input(shape=(n_features,))
hidden1 = Dense(64, activation="relu")(inpx)
drop1   = Dropout(0.3)(hidden1)
hidden2 = Dense(32, activation="relu")(drop1)
drop2   = Dropout(0.3)(hidden2)
out     = Dense(1, activation="sigmoid")(drop2)

model = Model(inputs=inpx, outputs=out)

model.compile(
    optimizer=Adam(learning_rate=0.001),
    loss=binary_crossentropy,
    metrics=["accuracy"]
)

model.summary()

history = model.fit(
    X_train, y_train,
    epochs=50,
    batch_size=128,
    validation_split=0.1,
    class_weight=class_weight,
    verbose=1
)
```

Bayesian Code

```
``` {r model_spec}
sigmoid <- function(z) 1 / (1 + exp(-z))

log_likelihood <- function(beta, X, y) {
  p <- sigmoid(X %*% beta)
  sum(y * log(p + 1e-10) + (1 - y) * log(1 - p + 1e-10))
}

log_prior <- function(beta, prior_sd = 10) {
  sum(dnorm(beta, mean = 0, sd = prior_sd, log = TRUE))
}
```
```

```
log_posterior <- function(beta, X, y) {
  log_likelihood(beta, X, y) + log_prior(beta)
}
```
```{r tuning_run}
metropolis_bayes <- function(X, y, n_iter = 5000, sigma = 0.01, burn = 0.18) {
  p <- ncol(X)
  beta_current <- rep(0, p)
  samples <- matrix(0, nrow = n_iter, ncol = p)
  accepted <- 0

  for (i in 1:n_iter) {
    beta_prop <- rnorm(p, mean = beta_current, sd = sigma)
    log_alpha <- log_posterior(beta_prop, X, y) - log_posterior(beta_current, X, y)

    if (log(runif(1)) < log_alpha) {
      beta_current <- beta_prop
      accepted <- accepted + 1
    }
    samples[i, ] <- beta_current
  }

  cat("Acceptance rate:", round(accepted / n_iter, 3), "\n")
  burn_n <- floor(burn * n_iter)
  return(samples[(burn_n + 1):n_iter, ])
}
```
```{r full_run}
set.seed(42)
samples <- metropolis_bayes(X_train, y_train, n_iter = 100000, sigma = 0.01,
                            burn = 0.18)

cat("Posterior samples shape:", nrow(samples), "x", ncol(samples), "\n")
```
```{r}
thinned_samples <- samples[seq(1, nrow(samples), by = 50), ]
colnames(thinned_samples) <- colnames(X_train)
cat("Thinned samples:", nrow(thinned_samples), "\n")
```

Works Cited

Rebollo, Juan Jose, and Hamsa Balakrishnan. 2014. “Characterization and Prediction of Air Traffic Delays.” Transportation Research Part C: Emerging Technologies 44: 231–241. https://doi.org/10.1016/j.trc.2014.04.007

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Qu, Jingyi, Ting Zhao, Meng Ye, Jiayi Li, and Chao Liu. 2020. “Flight Delay Prediction Using Deep Convolutional Neural Network Based on Fusion of Meteorological Data.” Neural Processing Letters 52: 1–19. https://doi.org/10.1007/s11063-020-10318-4